

# NITHISH KANNAN M

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## AI / ML Systems-Oriented Engineer (Undergraduate)

Chennai, Tamil Nadu, India

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## PROJECT & WORK EXPERIENCE

Academic & Independent Projects

**Vellore Institute of Technology, Chennai** | 2024 – Present

### UrbanPulse – Decision-Centric Intelligence for Urban Electricity Distribution (Ongoing)

- Built a regret-aware decision support system on urban electricity time-series logs.
- Focused on confidence-calibrated, explainable action recommendations over anomaly alerts.

### Adaptive Memory-Aware ML Inference Engine (Ongoing)

- Developed a runtime system that adapts ML inference under memory and latency constraints.
- Dynamically controlled precision and execution paths without retraining.

### Real-Time Student Crowd Management via Network Activity Analysis

- Designed a system to infer crowd density using network traffic patterns instead of camera-based vision.
- Applied clustering on network activity to identify dense regions and congestion behavior.

### Progressive NLP Pipeline with Attention-Based Sequence Modeling

- Implemented SVM → CNN-LSTM → Attention-LSTM pipelines for noisy text classification.
- Deployed inference via Flask REST API with reproducible training workflows.

## EXPERIENCE

- Deep Learning Engineer  
Artificial Intelligence Club, VIT Chennai
- UI/UX Designer  
Android Club, VIT Chennai

## EDUCATION

Vellore Institute of Technology (VIT), Chennai

B.Tech – Computer Science and Engineering (CORE)

**2nd Year** | CGPA: 8.6 / 10

## SKILLS

**Machine Learning & AI:** PyTorch, TensorFlow, Deep Learning, Attention Models, Time-Series Analysis

**Systems & Engineering:** Resource-Aware Inference, Model Quantization, Runtime Adaptation

**Programming:** Python, C, C++, Java

**Data & Visualization:** R (basic), NumPy, Matplotlib

**Tools & Platforms:** Git, Linux, Flask

**CERTIFICATION:** Machine Learning – Andrew Ng (Stanford University), GenAI IBM Professional (In Progress)

**INTERESTS:** Sports & Fitness | Entrepreneurship | AI Systems & ML Infrastructure